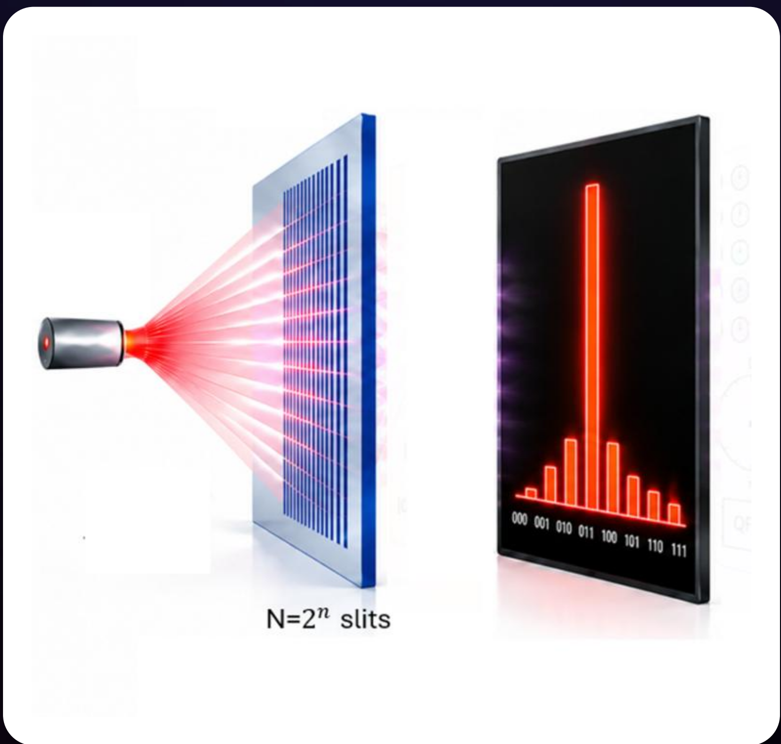
A VISUAL SERIES · PROJECT 2

# Quantum Computing, Across Disciplines

*. From the double-slit model to Quantum Computing*

One shared journey, read through five disciplinary lenses. We map the framework onto **Quantum Phase Estimation**, run it end to end, and re-read a published result through the same structure.



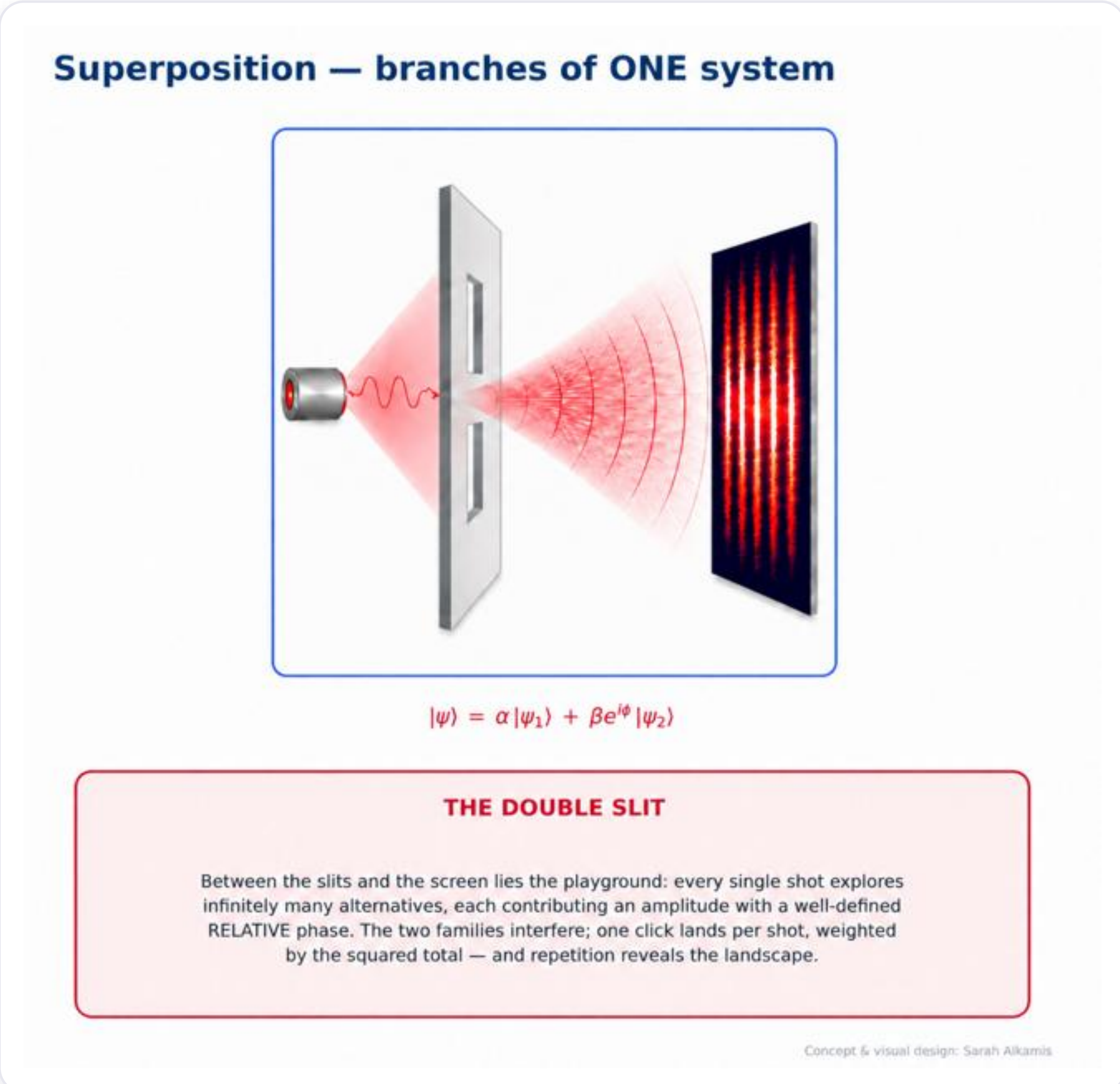

*Two paths → N = $2^n$ encoded alternatives · an abstract N-slit boundary object*


**Sarah Ibrahim Alkamis**

Concept & visual design

PROJECT 1 · RECAP

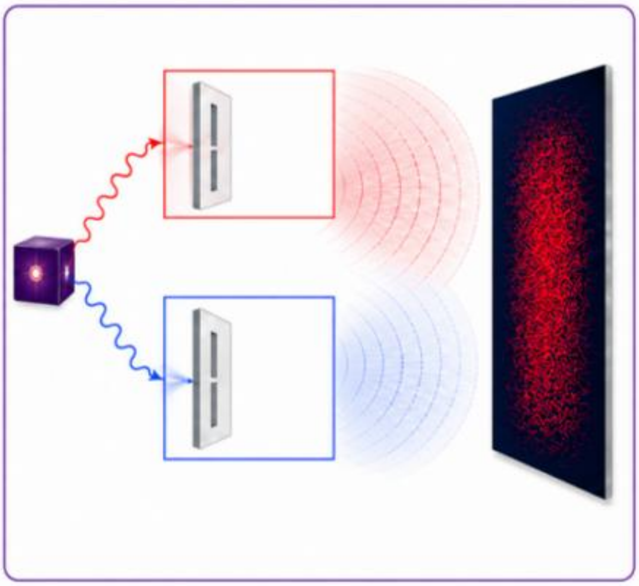
# Superposition: coherent branches of one system



PROJECT 1 · RECAP

# Entanglement: shared branches of a composite system

## Entanglement — shared branches of a COMPOSITE system



$$|\Psi\rangle_{AB} = \alpha\,|\psi_1^A\rangle|\psi_1^B\rangle + \beta e^{i\phi}\,|\psi_2^A\rangle|\psi_2^B\rangle$$

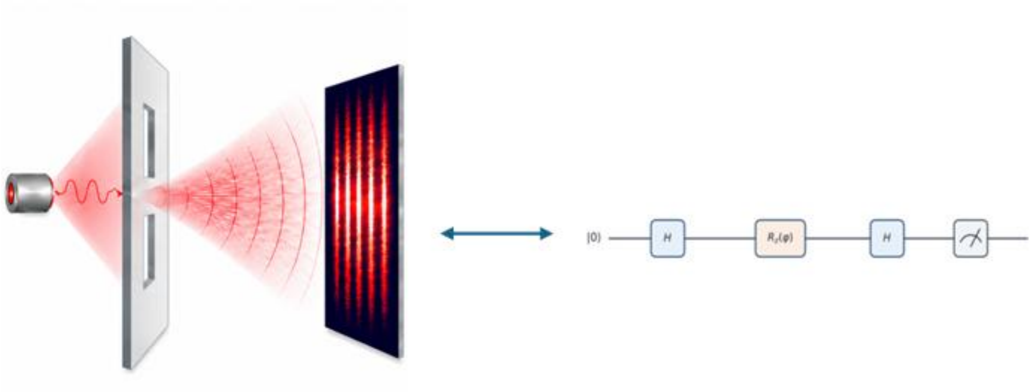
### THE SECOND-ORDER DOUBLE SLIT

One conservation law selects two grand slits for the pair — and inside each, every system is still a first-order double slit of its own. For a maximally entangled pair, which-path detection on either partner acts on the grand slits — no local fringes; the phase lives in the joint correlations.

Concept & visual design: Sarah Alkamis

THE FIRST BRIDGE

# One experiment, two languages

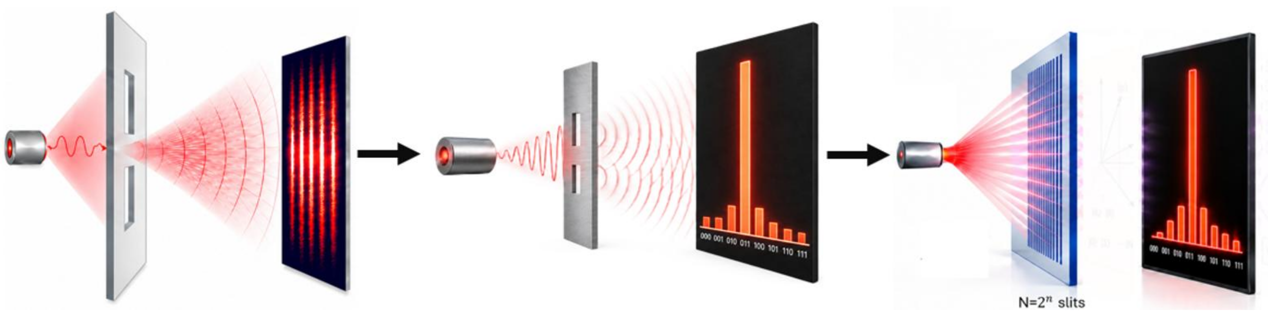



| PHYSICIST | STAGE | QC RESEARCHER |
|---|---|---|
| source emits one particle | **Prepare** | initialize $\|0\rangle$ |
| barrier with two slits | **Split** | Hadamard $H$ |
| path difference sets the angle $\Delta\varphi$ | **Phase** | dial the angle: $R_z(\varphi)$ |
| propagation to the screen | **Recombine** | second $H$ |
| click at position $x$ | **Detect** | measure bit 0 / 1 |

**The interferometer is a one-qubit circuit:** prepare |0>, split with a Hadamard, imprint the path phase with R_z(ϕ), recombine with a second Hadamard, then measure. Same amplitude mathematics, two languages.

THE ENCODING

# From optical modes to qubits



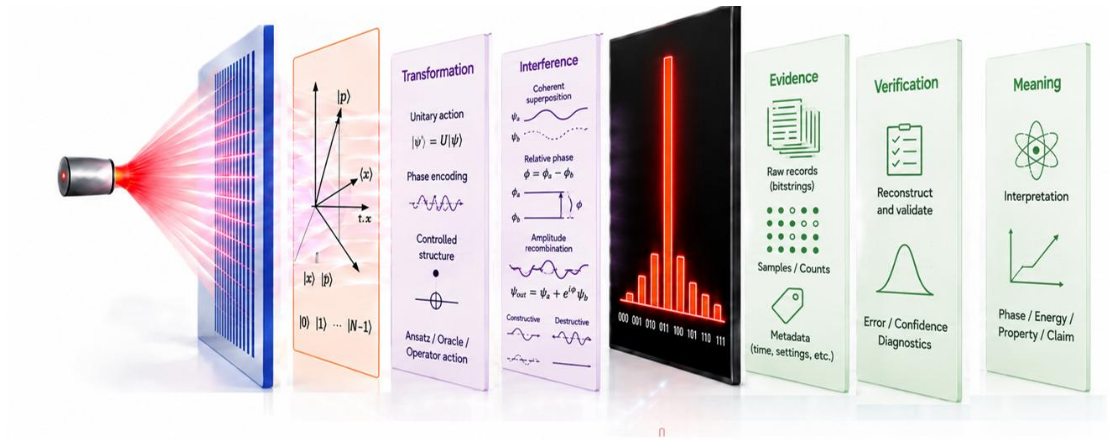

coherent alternatives | relative phase | amplitude recombination | probabilistic readout

**2 modes ↔ 1 qubit · $2^n$ modes ↔ n qubits**

**Not slits-as-qubits — a declared encoding.** The abstract N-slit model is a boundary object: a shared anchor tested across algorithm families, letting each discipline read the same algorithm in its own vocabulary.

THE PIPELINE

# One shared journey, eight stages




**1 Origin**
problem & initial state

**2 Representation**
encoding & basis

**3 Transformation**
unitary action; gates & operators

**4 Interference**
amplitude recombination; relative phase

**5 Measurement**
outcomes / bitstrings

**6 Evidence**
raw records, counts, metadata

**7 Verification**
reconstruct, compare, validate

**8 Meaning**
interpretation / the claim

THREE SUPPORTS

# One backbone, three supports


**1**

### Five analytical lenses

What forms of reasoning are active?

**2**

### Cross-cutting quality rail

How is trust built at every transition?

**3**

### Programmatic & execution path

How is the journey realized as executable artifacts?

**A scaffold makes the full field navigable before full mastery.**

THE LENSES

# Five lenses, not five people

The lenses describe analytical functions — not job titles or ownership.

1. **Domain & Physical** — what problem, and why it matters.

2. **Mathematical Modelling** — the formal structure underneath.

3. **Quantum Computation** — gates, circuits, and the quantum core.

4. **Classical Computation & Evidence** — records, estimators, statistics, validation.

5. **Execution & Hardware** — the physical device that runs it.

**A stage locates the work; a lens interrogates it.**

THE MAP

# Read the shape, not every cell


**Read centers of gravity and handoffs:** mathematics concentrates on representation; quantum reasoning on transformation and interference; classical evidence spans measurement to verification; hardware sits at execution; domain reasoning returns the result to meaning.

THE EXECUTION PATH

# From idea to executable evidence

1 **Build**
mission, object, and circuit

2 **Translate**
serialize & import the circuit artifact

3 **Validate**
test structural & semantic contracts

4 **Transpile**
adapt to the selected backend

5 **Execute**
submit the compiled circuit

6 **Preserve evidence**
retain records, settings & provenance

7 **Reconstruct**
decode the estimate & diagnostics

**Code does not replace the conceptual journey — it realizes it through a traceable chain of artifacts.**

THE QUALITY RAIL

# Verification is a stage. Quality is a rail.

Reconstruct, compare against a reference, quantify uncertainty, and test whether the evidence supports the claim.

- A — **Assumptions & approximations** — what is taken for granted.
- T — **Tests & edge cases** — where the method breaks.
- U — **Uncertainty & error budget** — how large is the residual doubt.
- P — **Provenance & reproducibility** — can the result be regenerated.
- I — **Interface contracts** — what each stage promises the next.

**At every point: what is the claim? what could fail? what check reveals it? what evidence must be kept?**

IN PRACTICE

# Two ways to use the framework

## Operational demonstration

Map and run **Quantum Phase Estimation** end to end. Follow the algorithm from the shared conceptual body into executable artifacts, preserve the raw evidence and provenance, reconstruct the phase estimate, and verify the final claim.

**Build → Translate → Transpile → Execute → Evidence → Reconstruct → Verify**

## Illustrative cross-lens example

Re-read a published result — the **ParityQC QFT case** — as a hardware/routing constraint addressed through representation and synthesis. The next slide unfolds it.

**See it → map it → run it → preserve the evidence → verify it → explain it.**

A WORKED CASE

# ParityQC: a cross-lens reading

*Klaver et al., "SWAP-less implementation of quantum algorithms," Phys. Rev. A 113, 012443 (2026).*

### Constraint · Hardware / routing

Limited connectivity and SWAP overhead: distant logical interactions become expensive.

### Intervention · Representation / synthesis

Re-represent information as parity labels; use entangling gates as information transport, not only state transformation.

### Reading · Across two lenses

A constraint visible at the hardware layer, addressed by a representation- and synthesis-level intervention.

Hardware / routing → Representation / synthesis

# See the whole journey.

- Look at quantum computing from the outside before diving in.
- Locate your discipline within the journey.
- See what the other disciplines contribute.
- Then run an algorithm as one connected workflow.

| Carousel | Guide | Colab |
| --- | --- | --- |
| the full architecture | the interdisciplinary foundation | map, run, inspect, verify |

Sarah Ibrahim Alkamis